

Evaluating the Security Posture of Default IaaS Configurations in Azure vs Google Cloud

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Abstract

This project aims to evaluate the security posture of default Infrastructure-as-a-Service (IaaS) configurations in Microsoft Azure and Google Cloud Platform (GCP). As organizations increasingly migrate to cloud platforms, many rely on default security settings without fully understanding the risks involved. We compare both platforms across key security dimensions including Identity and Access Management (IAM), storage security, infrastructure configurations, and logging/monitoring. Our goal is to identify vulnerabilities inherent in default settings, provide actionable recommendations, and guide organizations toward more secure cloud practices.

Introduction

Cloud adoption has grown significantly, increasing the need for robust security measures to protect sensitive assets. Microsoft Azure and Google Cloud Platform (GCP) offer strong encryption algorithms, identity management, and threat detection, yet their default IaaS configurations may still pose risks, including permissive access controls and weak encryption policies. This report evaluates the security posture of Azure and GCP's default configurations.

Objectives:

- Assess **infrastructure configurations** for potential vulnerabilities.
- Evaluate **storage security** policies and access controls.
- Compare **logging and monitoring** capabilities.
- Conduct **risk analysis** and provide actionable recommendations.

Methodology

To conduct a comparative security analysis of default configurations in Google Cloud Platform (GCP) and Microsoft Azure, we deployed a controlled test environment in both platforms using their default Infrastructure-as-a-Service (IaaS) settings. Our methodology included the following key steps:

- **Cloud Resource Deployment:** We provisioned cloud storage buckets and virtual machines on both platforms using default settings. The provisioning process was documented, and any differences—particularly those that could introduce security risks—were identified and analyzed.
- **Configuration Assessment:** We evaluated the default configurations of virtual machines and storage resources from a security perspective. This included examining aspects such as firewall rules, access control, encryption settings, and public exposure.
- **Security Tooling and Documentation Review:** We explored and compared the default logging and monitoring tools provided by each platform, including Microsoft Defender for Cloud and Google Security Command Center. Official documentation was reviewed to understand the scope and capabilities of each toolset.
- **Comparative Analysis:** All findings were systematically compared to assess each platform's approach to default security across infrastructure, storage, monitoring, and overall risk exposure.

Enhanced Security Evaluation Summary:

This extended methodology supplements our core evaluation by incorporating comparative security testing across all three major cloud platforms: AWS, Microsoft Azure, and Google Cloud Platform (GCP). For this stage, we provisioned VMs and storage instances using default configurations and examined them for inherent risks. Key evaluation criteria included IAM policies, public exposure, encryption defaults, logging status, and available threat detection tools. Each platform's support for industry-standard frameworks (like CIS Benchmarks and NIST guidelines) was considered to assess readiness against enterprise security standards.

Analysis:

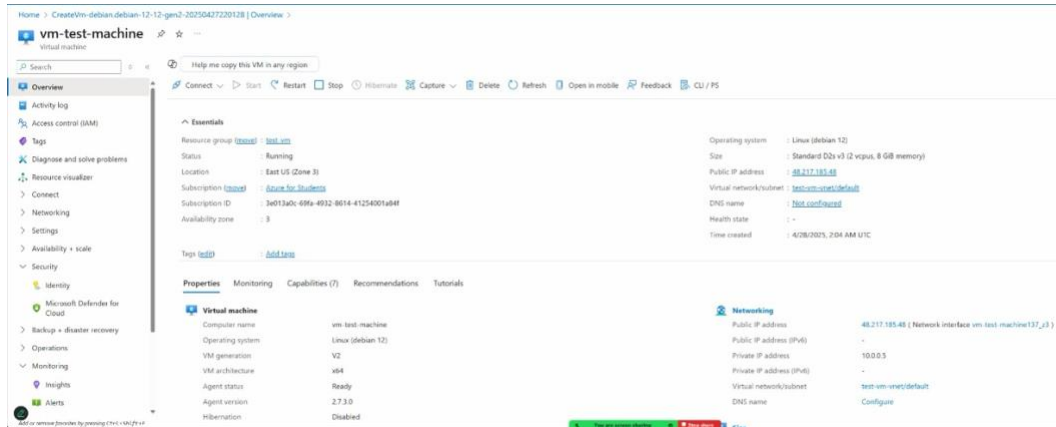
Our extended analysis revealed that while both Azure and GCP offer robust foundational security by default, they differ significantly in granularity of control and visibility features. Google Cloud Platform (GCP) stands out with object-level IAM controls, automatic encryption, and Access Transparency - offering fine-grained permissioning and auditability. On the other hand, Microsoft Azure provides deep integration with Azure Active Directory (AD) and supports layered encryption, including double encryption through Customer-Managed Keys layered over platform-managed keys. However, both platforms present security concerns in their default states: GCP's default service accounts often have broad project-level permissions, increasing the risk of privilege escalation, whereas Azure's default configurations typically have network flow logging disabled, potentially limiting visibility into traffic patterns and lateral movement detection.

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1. Virtual Machine Environments

Creating and managing virtual machines (VMs) is central to IaaS usage. Both Microsoft Azure and Google Cloud Platform (GCP) offer flexible deployment and configuration options with built-in security features. However, their default configurations differ in visibility, network isolation, and access management.

- **Azure Virtual Machines:** Azure VMs are provisioned via the Azure Portal, with configurations for OS image, VM size (e.g., B1s), authentication (SSH key or password), and network settings. While Azure offers built-in tools like Defender for Cloud and monitoring options, some security features—such as Network Security Group (NSG) logging and diagnostics—are disabled by default and require manual activation. Azure integrates tightly with Azure AD for identity-based access, and NSGs provide network isolation, though visibility gaps may arise without flow log activation.



- **GCP Compute Engine:** GCP VMs are created through the Google Cloud Console with customizable specs (e2-micro free tier is default). GCP supports browser-based SSH access and supports Shielded VMs to protect against rootkits and firmware tampering. IAM policies can be applied at the project or instance level. Like Azure, GCP's firewall rules and flow logs must be manually enabled for full visibility. Default service accounts are granted broad permissions, which may introduce privilege escalation risks.

Both Azure and GCP offer flexible VM creation workflows, but they require additional tuning to ensure secure, least-privilege deployments.

Key considerations include:

- **Access Control & Identity Risks:**
 - **Azure:** VM access via Azure AD and RBAC; however, default NSGs and VM monitoring are not fully enabled, potentially exposing gaps.
 - **GCP:** Service accounts assigned to VMs may have project-wide permissions by default, increasing risk if not scoped down.
- **Network Segmentation & Visibility:**
 - **Azure:** NSGs are used to isolate traffic, but flow logs must be manually activated.
 - **GCP:** VPC Firewall Rules are flexible, yet logging is also disabled by default, reducing immediate network transparency.

2. Infrastructure Configuration (IAM & Network Controls)

Identity and access management (IAM) and network segmentation form the foundation of a secure infrastructure. Azure and GCP both provide comprehensive identity platforms, but default settings may introduce unintended exposure.

- **Azure IAM & Network Controls:** Azure uses Azure AD in conjunction with Role-Based Access Control (RBAC), allowing role assignments at different scopes (subscription, resource group, VM). Managed Identities allow resources to authenticate without storing secrets. Network segmentation is

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handled via NSGs, but network logging is not enabled by default, potentially limiting visibility into traffic.

- **GCP IAM & Network Controls:** GCP's IAM supports role bindings at multiple resource levels—from organization down to individual objects—enabling a highly granular least-privilege approach. Network isolation is enforced using VPC firewall rules but logging and audit tracking are off by default. GCP also supports Workload Identity Federation for external identity integration without long-term key storage.

Default configurations in both platforms require customization to ensure proper access boundaries and audit visibility.

Key considerations include:

- **Privilege Escalation Risks:**
 - **Azure:** Over-permissioned default identities can grant broader access than necessary.
 - **GCP:** Default service accounts are assigned Editor or similar roles, increasing the attack surface.
- **Access Granularity:**
 - **Azure:** IAM granularity is available at the container or storage account level.
 - **GCP:** Supports object-level IAM and organization-wide policy enforcement.

3. Storage Security

Securing data at rest and enforcing proper access control are critical components of cloud storage strategy. While Azure and GCP offer strong default encryption, they differ in how access is managed and how granular the permissions can be.

- **Azure Storage Security:** Azure uses RBAC to control access at the storage account or container level. Object-level access is achieved using Shared Access Signatures (SAS), which provide time-limited and permission-scoped access. All data is encrypted at rest using AES-256, and HTTPS is enforced for in-transit encryption. Public access is enabled by default, thus needs immediate attention.

Basics		Data protection	
Subscription	Azure for Students	Point-in-time restore	Disabled
Resource group	test-vm	Blob soft delete	Enabled
Location	East US	Blob retainment period in days	7
Storage account name	teststore1232434	Container soft delete	Enabled
Primary service		Container retainment period in days	7
Performance	Standard	File share soft delete	Enabled
Replication	Read-access geo-redundant storage (RA-GRS)	File share retainment period in days	7
Advanced		Versioning	Disabled
Enable hierarchical namespace	Disabled	Blob change feed	Disabled
Enable SFTP	Disabled	Version-level immutability support	Disabled
Enable network file system v3	Disabled	Encryption	
Allow cross-tenant replication	Disabled	Encryption type	Microsoft-managed keys (MMK)
Access tier	Hot	Enable support for customer-managed keys	Blobs and files only
Enable large file shares	Enabled	Enable infrastructure encryption	Disabled
Security			
Secure transfer	Enabled		
Blob anonymous access	Disabled		
Allow storage account key access	Enabled		
Default to Microsoft Entra authorization in the Azure portal	Disabled		
Minimum TLS version	Version 1.2		
Permitted scope for copy operations (preview)	From any storage account		
Networking			
Network connectivity	Public endpoint (all networks)		
Default routing tier	Microsoft network routing		

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- **GCP Storage Security:** GCP supports IAM bindings down to the individual object level, enabling tighter access control. It also offers Signed URLs for temporary object access. By default, GCP encrypts all data at rest using AES-256 with Google-managed keys, and in-transit communication is protected via TLS 1.2 or higher. Public access is also disabled by default but can be changed at both the bucket and object level.

The screenshot displays the Google Cloud Storage bucket configuration interface. It includes several sections for configuring the bucket's security and access:

- Choose where to store your data:** Location: us (multiple regions in United States), Location type: Multi-region.
- Choose how to store your data:** Default storage class: Standard, Hierarchical namespace: Disabled.
- Choose how to control access to objects:** Public access prevention: On, Access control: Uniform.
- Choose how to protect object data:** Your data is always protected with Cloud Storage but you can also choose from these additional data protection options to add extra layers of security.
- Data protection:**
 - ☒ **Soft delete policy (For data recovery)**: When enabled, this bucket and its objects will be kept for a specified period after they're deleted and can be restored during this time. [Learn more](#)
 - ☒ **Use default retention duration**: All buckets have a 7 day soft delete duration by default, unless this default has been customized by your organization administrator.
 - ☐ **Set custom retention duration**: Specify how long this bucket and its objects should be retained after they're deleted. Setting a "0" duration disables soft delete, meaning any deleted objects will be permanently deleted.
 - ☐ **Object versioning (For version control)**: For restoring deleted or overwritten objects. To minimize the cost of storing versions, we recommend limiting the number of noncurrent versions per object and scheduling them to expire after a number of days. [Learn more](#)
 - ☐ **Retention (For compliance)**: For preventing the deletion or modification of the bucket's objects for a specified period of time.
- Data Encryption** (collapsed section)

At the bottom, there are "Create" and "Cancel" buttons.

Differences in access granularity and token-based sharing mechanisms reflect the platforms' unique design philosophies.

Key considerations include:

- **Access Control Granularity:**
 - **Azure:** RBAC applies at broader scopes; object-level access requires SAS tokens.
 - **GCP:** Full IAM control down to the object level; no need for separate token systems for fine-grained access.
- **Authentication & Access Models:**
 - **Azure:** Integrates deeply with Azure AD and supports Managed Identities.
 - **GCP:** Uses service accounts, JWTs, and Workload Identity Federation to eliminate the need for stored secrets.

4. Cryptography and Authentication Mechanisms

Encryption and secure authentication are foundational for protecting cloud workloads. Azure and GCP both enforce encryption by default and offer advanced key management options, but the implementations differ.

- **Azure Cryptography & Authentication:** Data at rest is encrypted with AES-256 and can use platform-managed, customer-managed (CMK), or customer-supplied keys (CSK). Azure supports double encryption (CMK + PMK) for sensitive data. Azure AD provides authentication with support for OAuth 2.0, OpenID Connect, and SAML 2.0. Managed Identities simplify service authentication without hardcoded secrets.

Encryption of data at rest

Data at rest includes information that resides in persistent storage on physical media, in any digital format. The media can include files on magnetic or optical media, archived data, and data backups. Microsoft Azure offers a variety of data storage solutions to meet different needs, including file, disk, blob, and table storage. Microsoft also provides encryption to protect [Azure SQL Database](#), [Azure Cosmos DB](#), and [Azure Data Lake](#).

Data encryption at rest using AES 256 data encryption is available for services across the software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS) cloud models. This article summarizes and provides resources to help you use the Azure encryption options.

For a more detailed discussion of how data at rest is encrypted in Azure, see [Azure Data Encryption-at-Rest](#).

Azure encryption models

Azure supports various encryption models, including server-side encryption that uses service-managed keys, customer-managed keys in Key Vault, or customer-managed keys on customer-controlled hardware. With client-side encryption, you can manage and store keys on-premises or in another secure location.

- **GCP Cryptography & Authentication:** GCP also uses AES-256 encryption by default with Google-managed keys, while allowing users to configure CMEK or CSEK for tighter control. GCP supports strong authentication mechanisms including service accounts, JWTs, OAuth 2.0, and Workload Identity Federation to securely authenticate workloads across environments.

Default encryption at rest

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This content was last updated in May 2024 and represents the status quo as of the time that it was written. Google's security policies and systems may change going forward, as we continually improve protection for our customers.

At Google, our comprehensive security strategy includes encryption at rest, which helps to protect customer data from attackers. We encrypt all Google customer content at rest, without any action required by you, using one or more encryption mechanisms. This document describes our approach to default encryption at rest for Google infrastructure and Google Cloud, and how we use it to keep customer content more secure.

This document is for security architects and security teams who are currently using or considering Google. This document assumes a basic understanding of [encryption](#) and [cryptographic primitives](#). For more information on cryptography, see [Introduction to Modern Cryptography](#).

Encryption at rest is encryption that is used to help protect data that is stored on a disk (including solid-state drives) or backup media. All data that is stored by Google is encrypted at the storage layer using the Advanced Encryption Standard (AES) algorithm, AES-256. We use a common cryptographic library, Tink, which includes our FIPS 140-2 validated module (named [BoringCrypto](#)) to implement encryption consistently across Google Cloud.

We own and manage the keys used in default encryption at rest. If you use Google Cloud, Cloud Key Management Service lets you create your own encryption keys that you can use to add envelope encryption to your data. Using Cloud KMS, you can create, rotate, track, and delete keys. For more information, see [Cloud Key Management Service deep dive](#).

Both platforms meet encryption standards but differ in visibility and customer control over key lifecycle.

Key considerations include:

- **Encryption Control & Compliance:**
 - **Azure:** Double encryption is supported but not enabled by default.
 - **GCP:** CMEK and CSEK must be manually configured for customer control.
- **Authentication Models:**
 - **Azure:** Primarily relies on AD integration and Managed Identities.
 - **GCP:** Supports secure, ephemeral identities via Workload Identity Federation.

5. Key Management

Key management determines how encryption keys are created, stored, and accessed. Both Azure and GCP provide strong solutions with support for hardware-backed protection and customer-defined lifecycle policies.

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- **Azure Key Management:** Azure Key Vault centralizes key and secret management, supporting platform-managed keys (PMK), customer-managed keys (CMK), and bring-your-own-key (BYOK) models. It also supports dedicated HSMs compliant with FIPS 140-2. Integration with Azure Policy and Monitor enables complete auditing and fine-grained RBAC-based access control.
- **GCP Key Management:** GCP offers Cloud KMS and Cloud HSM to manage software- and hardware-backed keys. Default encryption uses Google-managed keys, but users can configure CMEK for compliance and CSEK for enhanced control. Access policies are enforced through IAM, with complete audit logging and support for key rotation and versioning.

Despite similar support for key customization and HSMs, differences remain in visibility and ease of integration.

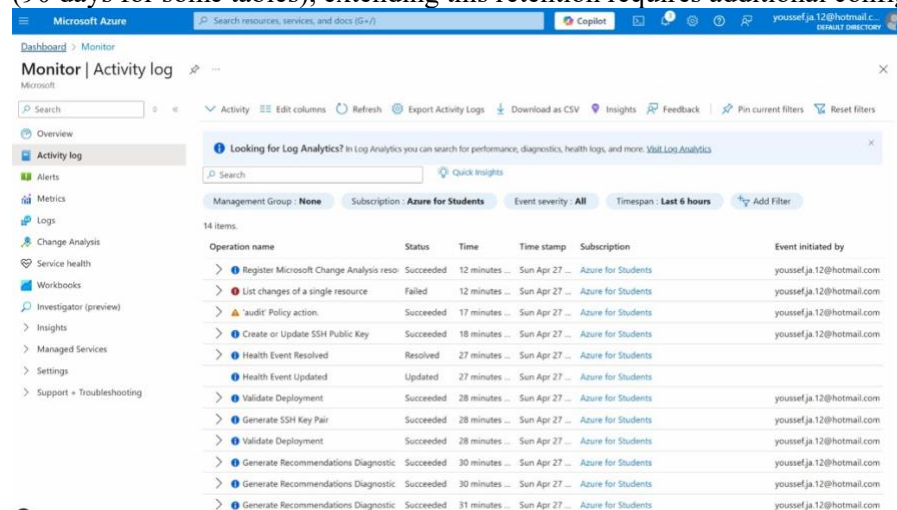
Key considerations include:

- **Visibility & Auditability:**
 - **Azure:** Offers detailed access logs and integrates with Sentinel and Azure Policy. Azure also provides better transparency into the key management processes.
 - **GCP:** Enables audit logging but requires explicit configuration for granular tracking.
- **Deployment Models:**
 - **Azure:** Supports PMK, CMK, CSK, BYOK, and HYOK.
 - **GCP:** Supports CMEK, CSEK, and BYOK, but not HYOK natively.

6. Logging & Monitoring:

Effective logging and monitoring are critical components of a secure cloud environment. Both Microsoft Azure and Google Cloud Platform (GCP) provide robust toolsets for visibility and threat detection, but they differ in default configurations, integration depth, and retention capabilities.

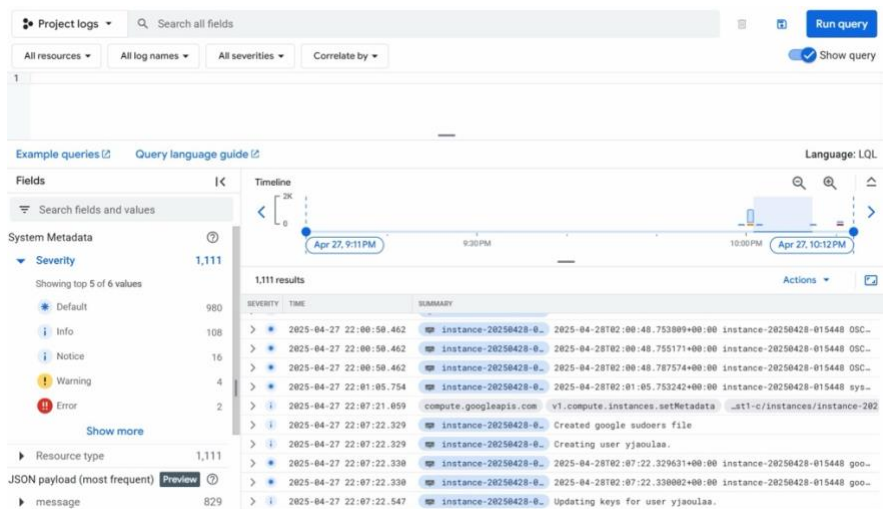
Azure Logging & Monitoring: Azure provides a comprehensive suite including Azure Monitor, Activity Logs, Diagnostic Logs, Microsoft Defender for Cloud, and Azure Sentinel. These tools collectively offer extensive monitoring of applications, infrastructure, and network resources. However, some capabilities—such as NSG (Network Security Group) Flow Logs and Firewall Logs—must be manually enabled, which introduces potential visibility gaps in default configurations. Although Azure offers free 30-day retention (90 days for some tables), extending this retention requires additional configuration and storage costs.



Operation name	Status	Time	Time stamp	Subscription	Event initiated by
Register Microsoft Change Analysis reso	Succeeded	12 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
List changes of a single resource	Failed	12 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
audit Policy action	Succeeded	17 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Create or Update SSH Public Key	Succeeded	18 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Health Event Resolved	Resolved	27 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Health Event Updated	Updated	27 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Validate Deployment	Succeeded	28 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Generate SSH Key Pair	Succeeded	28 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Validate Deployment	Succeeded	28 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Generate Recommendations Diagnostic	Succeeded	30 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Generate Recommendations Diagnostic	Succeeded	30 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com
Generate Recommendations Diagnostic	Succeeded	31 minutes ...	Sun Apr 27 ...	Azure for Students	youssefja.12@hotmail.com

GCP Logging & Monitoring: GCP's ecosystem includes Cloud Monitoring, Cloud Logging, Cloud Audit Logs, and Security Command Center (SCC). These tools provide centralized visibility with automatic integration across services. GCP offers free log storage for up to 365 days if logs are moved to Cloud Storage, giving it a cost-effective edge in long-term retention. However, VPC Flow Logs, Firewall Rule Logs, and API data access logs are disabled by default, requiring manual intervention.

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Both Azure and GCP present security risks in their default logging and monitoring configurations, especially when it comes to network visibility and threat detection. Key considerations include:

- **Network Visibility Gaps:**
 - **Azure:** NSG Flow Logs and firewall logging are disabled by default, which may delay detection of unauthorized access or lateral movement.
 - **GCP:** VPC Flow Logs, Firewall Rules Logging, and Data Access Audit Logs are also disabled by default, requiring manual activation to ensure visibility into key activities.
- **Alerting Limitations:**
 - Both platforms offer alerting features, but default thresholds are often too broad.
 - This can lead to missed threats or an overload of false positives, reducing the effectiveness of real-time monitoring.

Set up recommended alert rules

Select alert rules

>	Percentage CPU is greater than	80	%	☒
>	Available Memory Bytes is less than	1	GB	☒
>	Data Disk IOPS Consumed Percentage is greater than	95	%	☒
>	OS Disk IOPS Consumed Percentage is greater than	95	%	☒
>	Network In Total is greater than	500	GB	☒
>	Network Out Total is greater than	200	GB	☒
>	VmAvailabilityMetric is less than	1		☒

- **Log Retention and Compliance Risks:**
 - **Azure:** Offers up to 90 days of log retention for certain services.
 - **GCP:** Provides 30 days by default, but allows free long-term storage in Cloud Storage with manual setup.
 - Inadequate retention can result in compliance issues (e.g., PCI-DSS, HIPAA) or insufficient forensic data after an incident.
- **SIEM Integration Differences:**

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- **Azure:** Includes Microsoft Sentinel, a native SIEM with deep integration into its ecosystem.
- **GCP:** Relies more on third-party solutions like Chronicle, which may require additional setup time and resources.

Recommendations

Below are some key recommendations to follow when provisioning and managing infrastructure in the cloud:

- Implement Principle of Least Privilege
 - Customize IAM roles and permissions to minimize privilege escalation risks.
 - Regularly review and limit IAM roles and policies to enforce least privilege access.
- Enhance Visibility and Threat Detection
 - Enable critical logging features, including network flow logs, firewall logs, and API access logs.
 - Utilize integrated or external SIEM tools (e.g., Azure Sentinel, Chronicle, GuardDuty) for continuous monitoring.
 - Customize SIEM rules and alert thresholds to align with the organization's threat model and operational priorities.
- Secure Data with Strong Encryption Practices
 - Activate Customer-Managed Encryption Keys (CMKs) to maintain control over key lifecycle.
 - Enforce automated key rotation policies to meet compliance and security requirements.
- Harden Infrastructure and Network Configuration
 - Customize infrastructure security settings and apply network segmentation to reduce lateral movement.
 - Use Infrastructure as Code (IaC) tools (e.g., Terraform, Bicep) to deploy consistent and secure environments rather than relying on manual console configurations.
- Maintain Compliance and Apply Best Practices
 - Apply established security baselines (e.g., CIS Benchmarks) for virtual machines, storage, and networking.
 - Conduct regular compliance checks against relevant regulatory standards such as HIPAA, GDPR, and FedRAMP.

Conclusion

This study reinforces the critical realization that default configurations in cloud platforms like Azure and Google Cloud are merely foundational and not sufficient for enterprise-grade security. While both platforms offer robust tool sets, such as identity and access management, encryption, and monitoring. The default states often leave critical gaps in areas like key control, network visibility, and privilege management.

Through hands-on evaluation and documentation review, we identified significant security variances between Azure and GCP in how they approach access control, encryption enforcement, key storage, and logging. Azure's deep integration with Active Directory and layered encryption provides advantages in hybrid enterprise settings, while GCP's object-level IAM and default transparency features offer superior control and auditability for modern workloads.

Ultimately, organizations must move beyond relying on "secure by default" assumptions. Custom IAM roles, encryption key policies, and proactive logging strategies are essential to minimizing risk and achieving compliance. By applying tailored security enhancements and aligning configurations with regulatory and operational needs, cloud adopters can significantly reduce their threat surface and build a resilient cloud infrastructure that supports innovation without compromising on security.

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